COM105 Engineering Physics (I)

Tutorial 10

1. Calculate A constant-volume gas thermometer registers an absolute pressure corresponding to 325 mm of mercury when in contact with water at the triple point. What pressure does it read when in contact with water at the normal boiling point?

2. The Humber Bridge in England has the world's longest single span, 1410 m. Calculate the change in length of the steel deck of the span when the temperature increases from -5.0 °C to 18.0 °C.

3. A U.S. penny has a diameter of 1.9000 cm at 20.0 °C. The coin is made of a metal alloy (mostly zinc) for which the coefficient of linear expansion is $2.6 \times 10^{-5}\ K^{-1}$. What would its diameter be on a hot day in Death Valley (48.0 °C)? On a cold night in the mountains of Greenland (-53 °C)?

4. A steel tank is completely filled with 1.90 m^3 of ethanol when both the tank and the ethanol are at 32.0 °C. When the tank and its contents have cooled to 18.0 °C, what additional volume of ethanol can be put into the tank?

5. A brass rod is 185 cm long and 1.60 cm in diameter. What force must be applied to each end of the rod to prevent it from contracting when it is cooled from 120.0 °C to 10.0 °C?

6. An aluminium tea kettle with mass 1.10 kg and containing 1.80 kg of water is placed on a stove. If no heat is lost to the surroundings, how much heat must be added to raise the temperature from 20.0 °C to 85.0 °C?

7. You are given a sample of metal and asked to determine its specific heat. You weigh the sample and find that its weight is 28.4 N. You carefully add 1.25×10^4 J of heat energy to the sample and find that its temperature rises 18.0 °C. What is the sample's specific heat?

8. A nail driven into a board increases in temperature. If we assume that 60% of the kinetic energy delivered by a 1.80-kg hammer with a speed of 7.80 m/s is transformed into heat that flows into the nail and does not flow out, what is the temperature increase of an 8.00-g aluminium nail after it is struck ten times?

9. A blacksmith cools a 1.20-kg chunk of iron, initially at 650.0 °C, by trickling 15.0 °C water over it. All of the water boils away, and the iron ends up at 120.0 °C. How much water did the blacksmith trickle over the iron? Given that the specific heat is 470 J/kg K for iron and 4190 J/kg K for water, while the specific latent heat of vaporization of water is 2256 kJ/kg.

10. In a container of negligible mass, 0.200 kg of ice at an initial temperature of -40.0 °C is mixed with a mass m of water that has an initial temperature of 80.0 °C. No heat is lost to the surroundings. If the final temperature of the system is 28.0 °C, what is the mass m of the water that was initially at 80.0 °C? Given that the specific heat is 2100 J/kg K for ice and 4190 J/kg K for water, while the specific latent heat of fusion of water is 334 kJ/kg.

11. Suppose that the rod in Fig. 17.24a is made of copper, is 45.0 cm long, and has a cross-sectional area of 1.25 cm^2. Let T_H = 100.0C and T_C = 0.0C. (a) What is the final steady-state temperature gradient along the rod? (b) What is the heat current in the rod in the final steady state? (c) What is the final steady-state temperature at a point in the rod 12.0 cm from its left end?

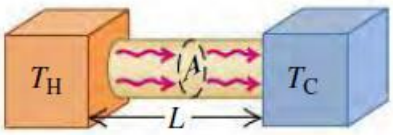
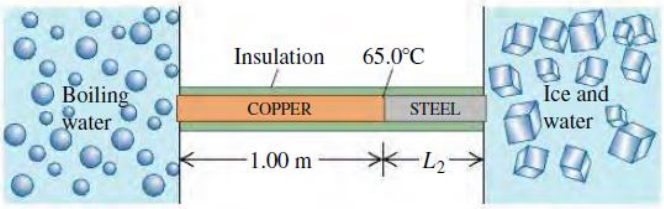


Fig. 17.24a

12. A long rod, insulated to prevent heat loss along its sides, is in perfect thermal contact with boiling water (at atmospheric pressure) at one end and with an ice–water mixture at the other (Fig. E17.62). The rod consists of a 1.00-m section of copper (one end in boiling water) joined end to end to a length L_2 of steel (one end in the ice–water mixture). Both sections of the rod have cross-sectional areas of 4.00 cm^2. The temperature of the copper–steel junction is 65.0 °C after a steady state has been set up. (a) How much heat per second flows from the boiling water to the ice– water mixture? (b) What is the length L_2 of the steel section?

Figure **E17.62**



13. A spherical pot contains 0.75 L of hot coffee (essentially water) at an initial temperature of 95 °C. The pot has an emissivity of 0.60, and the surroundings are at 20.0 °C. Calculate the coffee's rate of heat loss by radiation.